REACTIVE: Synergizing Active Sensing, Reasoning, and Acting in Language Models

Abstract

The name is read as *Re + Act + ively*: the first two halves are ReAct's loop, and the third half is what makes the loop start from the environment instead of from an instruction. ReAct interleaves reasoning traces with actions and lets a language model gather evidence before committing to an answer. Its loop, however, starts from a task someone else has already posed: a question, an instruction, a benchmark item. A personal agent faces an earlier and harder problem—most of the time there is no task at all, only a stream of low-level events, and the agent must decide whether any of them deserves to become one. We call this the *pre-task* problem and present REACTIVE, a loop that adds an *active sensing* stage in front of reasoning and acting. REACTIVE combines (i) a local publish/subscribe signal mesh in which any program can be a sensor, (ii) an *admission* gate that scores each signal on salience, urgency and quietness and returns one of four verdicts rather than a binary, (iii) an *experience* library of conditional rules carrying an explicit trust level, and (iv) *receipts*—a three-state artifact (done / waiting / held back) that makes silence auditable. We implement REACTIVE on a desktop agent runtime and report a case study on a single deployment. With one hundred background signals the system opened zero tickets; the same archiving rule executed in 13 seconds at trust level L1 and correctly refused to move a file at L3, where consent could not be obtained because the signal source had no reply channel. We document nine engineering failures encountered on the way, each of which changed the design.

1 Introduction

Reasoning and acting have usually been studied apart. Chain-of-thought prompting improves reasoning but stays inside a single forward pass, grounded only in what the model already memorized; acting agents such as WebGPT or SayCan interact with an environment but delegate the "why" to the policy. ReAct [1] unified the two: the model writes a thought, takes an action, reads an observation, and repeats, so that reasoning is grounded in retrieved evidence and actions are chosen for stated reasons.

The unification is real, but it preserves an assumption from its benchmarks: *the task is given*. HotpotQA supplies a question, ALFWorld supplies an instruction, WebShop supplies a shopping goal. Every ReAct trajectory begins after someone has decided that this episode should exist.

A personal agent is deployed before that decision is made. It runs while its user works, and the honest prior is that nothing needs to happen. Files land in a download folder; a clipboard changes; a sensor drifts. Almost all of it is noise, and the cost of being wrong is asymmetric: interrupting someone for nothing is far worse than staying silent. The interesting question is therefore not "how do I solve this task" but "*is there a task here, and should it be mine?*"

We call this the pre-task problem and give three reasons why the ReAct loop does not solve it.

No permission-free sensing. An agent that reasons over tool calls can only see what a tool returns, and a tool call is itself an action within a task. Continuous observation has no place in the loop: there is no step at which the model may look without already being busy.

Every observation costs a model call. Routing all events to the model makes silence the most expensive state. In our deployment a single mis-configured sensor produced 276 events in under two minutes; because each event triggered an assessment, the episode consumed more than 600K tokens before the storm was stopped. A personal agent must be cheap precisely when nothing is happening.

Asking is undefined. ReAct assumes the environment answers. In a proactive setting the agent often needs to ask the *person*, and there may be no channel to do so: if the signal arrived from a local sensor, the source has an address but no inbox. We found that an agent in this position does the right thing—it stops—but the stop is indistinguishable from a failure unless it is recorded as such.

REACTIVE addresses these with four additions to the ReAct loop: a sensing stage that runs continuously and independently of any task; an admission gate whose default verdict is silence; an experience library that encodes *when* a class of signals is worth acting on and *how much authority* the agent has; and receipts that make every decision, including the decision to do nothing, inspectable.

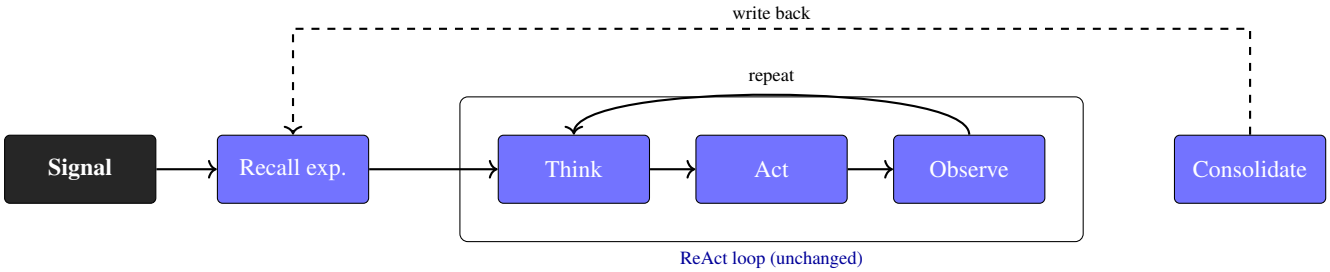


Figure 1: The REACTIVE paradigm: *Re* + *Act* + *ively*. ReAct supplies the first two halves—an inner think–act–observe loop; the third half is that the loop is entered by the world, not by an instruction. An experience is recalled before reasoning begins and is revised after acting, so the same situation is not reasoned about twice.

Our contributions are:

- A formulation of the pre-task problem and a four-stage loop (Sense → Admit → Recall → Act) that extends ReAct to continuously sensed environments (§2).
- An admission design in which the unit of output is a verdict on a spectrum (drop / record / digest / interrupt) rather than a binary, and in which a signal covered by no experience is inadmissible by default.
- The notion of an *executable* experience—condition, action, limits and trust level—and an empirical demonstration that vague rule text costs an order of magnitude more agent effort than explicit text.
- An implementation on a desktop agent runtime and a case study quantifying silence, trust-level ablation, and a real event storm (§4).

2 REACTIVE

2.1 Active sensing

Signals are not tool results. We model them as messages on a local publish/subscribe mesh in which every program can be a node: a file watcher, a clipboard monitor, a weather poller, a build daemon. Discovery is through a shared directory of leases; a node announces itself, peers connect, and `publish(path, payload)` broadcasts to all of them. There is no subscription table and no schema negotiation: the receiver filters locally by pattern, and a publisher neither knows nor needs to know who is listening.

This matters for two reasons. First, sensing becomes a property of the machine rather than of the agent: adding a sensor does not require editing the agent, and removing one does not break it. Second, and unlike a tool call, publishing is not an action *within* a task, so the sensing stage can run while the agent is idle—which is the normal case.

The design choice against polling is deliberate. A polling source samples the world on a fixed horizon; on a 60-second horizon, half of all events are reported late by more than 30 seconds, and the agent pays for the poll whether or not anything happened. Push inverts both: latency is bounded by the sensor, and an idle agent costs nothing.

Relation to tool protocols. A tool protocol such as MCP standardizes how an agent *calls* a program; the sensing stage standardizes how a program *notifies* the agent. The two are peers, not layers: one governs outbound actions, the other inbound observations. Screen understanding is a different problem—it reconstructs state from pixels, whereas sensing receives declared events.

2.2 Admission

The gate consults the experience layer first: an event that no experience covers is rejected before any scoring. Admission is the stage that decides whether a signal becomes a task, so it is drawn after recall in Figure 1. We score each signal by three factors:

$$s \;=\; \mathrm{base}(e) \;\times\; \mathrm{urgency}(e) \;\times\; \mathrm{quiet}(h), \tag{1}$$

where base reflects the event type's prior salience, $\mathrm{urgency}$ its time sensitivity, and $\mathrm{quiet}(h)$ a discount from recent interaction history h—an agent that has just spoken should be less willing to speak again.

kind	what it is	example	what it must have
Memory	what happened	"last Wednesday I downloaded a PDF called Invoice"	no action, no condition
Skill	what one can do	"I can move a file with PowerShell"	an action, no condition
Experience	what to do in which situation	"if takeout is over 40 minutes late, call and ask"	**condition + action + rationale + authority**

Table 1: Memory records, skills execute, experiences *decide*. Only an experience answers "when" and "with what authority" at once, which is why the trust levels attach to it and to nothing else.

The score maps to one of four verdicts: DROP (discard, count it), RECORD (keep for the day's digest, do not act), DIGEST (merge into a summary delivered later), and INTERRUPT (wake the agent now). The four-way output is the point. A binary decision forces every borderline event into a costly branch; a spectrum lets the system say "this is real but not now", which is what most real signals deserve.

Two properties of our gate are worth stating. First, admission is where *policy* lives, not the model: the model never sees a signal that the gate dropped, so a noisy sensor cannot inflate cost. Second, the gate is where the addressee problem appears—a verdict of "interrupt" presupposes somewhere to interrupt *to*.

2.3 Experiences

Memory, skill and experience are routinely conflated, so it is worth stating the difference operationally (Table 1). Memory records what happened and carries neither an action nor a condition; a skill carries an action but no condition, so it answers "how" and never "whether now". An experience carries both, plus a rationale and an authority level, and is therefore the only one of the three to which admission and trust can attach. The relation is that of ingredients, cookery and recipe: a recipe is derived from the first two and is neither. Two tests separate them: remove the condition and an experience collapses into a bare command, while a skill survives; remove the action and an experience is no longer an experience, while a memory is unaffected. A third test is decisive for our design: memories and skills need no permission, an experience does.

An experience is a conditional rule that a person teaches once and the agent applies thereafter:

field	content
condition	`file.extension in ["pdf","docx"]`
action	what to do, *with* the mechanism and the limits
trust	L1 act, L2 act and notify, L3 ask first, L4 report only
rationale	why, in one line

The action field is the crux, and our strongest empirical finding is about its wording. An action written as an *intention* ("pop a lightweight notification") leaves the mechanism unspecified, so the agent invents one: it searches the file system for a notifier, tries the messaging CLI, and finally writes and debugs a PowerShell toast script, spending 48 seconds and dozens of tool calls. Rewritten as an *instruction* ("do three things: check, move to `~/Documents/Work/`, report one line; do not read unrelated directories"), the same task completes in 13 seconds with three tool calls. We therefore treat executability as a first-class property of an experience and enforce it in the tool description through which the agent writes experiences—including a negative example, because models follow concrete contrasts far better than abstract requirements.

The experience layer is the least established part of REACTIVE. We treat it as a hypothesis under test rather than as a result: memory and skills can be summarized into a rule offline, but whether such a rule transfers across contexts is not settled by our data.

Experiences also supply the default-deny property. A signal that no experience covers is not admissible; the agent does not need a rule saying "do not handle weather". Silence is the absence of a rule, not the presence of a suppression.

2.4 Trust and the addressee gap

Trust levels convert "can the agent do this" from a prompt-time judgment into a stored, auditable fact. The same signal, the same experience and the same agent behave differently depending only on the stored level: at L3 the file is left untouched and a ticket is opened; at L1 the file is moved and a one-line report is produced.

L3 exposes a structural gap. Asking presupposes a channel, and a locally sensed event has none: the source is a sensor, not a conversation. Our agent discovered this itself and reported it in exactly these terms:

"the confirmation question cannot be delivered—the MSP source (`msp:file.created`) has no reply channel"

It then declined to move the file, searched for an alternative channel, read the source-kind interface to confirm that the channel was optional and unimplemented, and recorded the ticket as blocked. This is the correct behaviour, and it is also a design requirement: an agent whose trust model includes "ask first" must be given somewhere to ask.

2.5 Receipts

The durable artifact of a REACTIVE episode is not a chat transcript but a receipt with three states: *done* (what was done, where the result is), *waiting* (one question, with a default), and *held back* (why nothing was done). The third state is the one that carries the most information: it is the only evidence that the system saw something and chose restraint. Our primary metric is the *intervention rate*—the fraction of sensed signals that reach the person—because it is the only measure that is honest about a proactive system's cost to attention.

3 Implementation

We implement REACTIVE as a plugin bundle for a desktop agent runtime (Cordis-based), with the signal mesh written in TypeScript and vendored into the bundle so the plugin has no external dependency. The sensing stage is a set of source kinds; a source declares its paths, its scope (listened / observed / addressed) and, optionally, a reply channel. A file-watching source is itself packaged as a plugin and a mesh node at once: it watches the direct children of a directory, waits for a file's size to settle, and publishes `file.created` with {`name, path, bytes`}.

Two implementation details proved essential. First, deduplication cannot be delegated to the runtime: the runtime neither rate-limits nor deduplicates, because remote sources such as a code host are assumed to be well behaved. A local sensor is not, so the source deduplicates on `(path, name, bytes)` within a time window. Second, an experience library must be re-read per signal rather than cached, otherwise a rule taught in conversation does not affect the next event.

4 Case study

Setup. One user, one desktop (Windows 11), one agent runtime, three sensors: a file watcher over the download folder, a clipboard monitor, and a heartbeat. The experience library contained three rules (archive work documents; ask before installing; keep videos in place). We report a single deployment and do not claim statistical generality; our purpose is to exhibit the loop's behaviours and their costs.

Silence is the default. We injected one hundred background signals (heartbeat ticks, clipboard changes, file events outside any experienced class). The system opened zero tickets and spent its budget on admission rather than on acting; the model was consulted only for events that reached the gate, and every such event was recorded with a one-line reason. The intervention rate was 0%.

Trust levels change behaviour, not capability. The same archiving rule, applied to three files, produced (Table 2):

file	trust	outcome	note
`Spec-2026.pdf`	L3	blocked, file untouched	consent unreachable
`Design-2027.pdf`	L1	moved in 13 s	one-line report
`Brief-2028.pdf`	L1	moved in 13 s	one-line report

Table 2: Same signal, same experience, same agent; only the stored trust level differs.

Experiences must be found to be useful. Instrumenting the pipeline showed a PDF event matching one of three experiences (`exp 1/3`) and a clipboard event matching none. With no matching experience the clipboard event was ignored with the reason "no experience covers it" rather than acted upon—the default-deny property in operation.

Vagueness is expensive. The first version of the archiving rule said only "pop a lightweight notification". The agent's trajectory ran to 48 seconds and dozens of tool calls, including a search for the desktop notification database. The rewritten, executable version completed in 13 seconds. The cost difference is not in the model but in the specification.

An event storm. A defect in an early watcher (a state flag re-enqueued the same file) emitted the same three events every 1.5 seconds. The runtime ingested 276 events and spent more than 600K tokens before the sensor was stopped by hand. Two fixes followed: a state machine in the sensor (idle / queued / emitted) and a deduplication window in the source. Neither is a model-side fix; both are consequences of treating sensing as a first-class stage rather than as a series of tool calls.

Engineering failures that changed the design. Nine defects were found and fixed during deployment: five were POSIX assumptions in a runtime targeting Windows (an `env` wrapper for credential probes, a mandatory repository, an unset state directory that failed silently, a shell name hard-coded to `bash`, and a YAML document-separator collision when seeding a profile); two belonged to the mesh (a wildcard pattern that matched nothing, and a package-resolution assumption that broke on Windows); two were ours (a mesh node that was never stopped across plugin reloads, and a lock left behind after a forced kill). Each is recorded with its symptom and fix in the project's issue log.

5 Related work

ReAct [1] interleaves reasoning and acting for task-directed agents; REACTIVE keeps that loop and adds the stages that precede a task. Toolformer [2] and its successors teach models to call tools; the sensing stage differs in that publishing a signal is not a tool call and needs no model decision. Voyager [3] accumulates a skill library in an embodied setting; our experience library differs in storing *conditions and authority* rather than code, and in being written in conversation by the user. Generative agents [4] simulate believable behaviour with memory streams; REACTIVE targets deployment, where the operative question is whether to act at all. Proactive assistants [5] predict when to intervene from interaction history; we add an explicit, inspectable gate and a trust level that bounds what an intervention may do. Model Context Protocol standardizes tool access for a model; our mesh standardizes *observation* across programs that were not written to be observed.

6 Limitations

Our evidence is a single-user, single-machine case study with no baseline system and no human evaluation. The intervention rate is reported for injected signals, not for organic use over weeks, and the trust-level comparison involves three files. The cost figures are dominated by the storm we describe and should not be read as steady-state numbers. The addressee gap is diagnosed but not yet closed: our implementation still has no local reply channel, so L3 tickets queue as blocked rather than being answered in place. Finally, admission currently uses hand-set weights; learning them from receipt history is left to future work.

7 Conclusion

ReAct showed that a language model can reason about what to do next. A personal agent must first decide whether to do anything at all. REACTIVE adds that decision—sensing, admission, recall of experience, and trust-bounded action—and makes the result auditable through receipts whose most informative state is "held back". In our deployment the loop stayed silent on one hundred background signals, executed an experienced task in thirteen seconds at one trust level, and refused the same task at another because it had no way to ask. The refusal is the more interesting result: it shows that the limiting factor for proactive agents is not capability but *addressability*.

References

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